

	Case Name: Investment Product (1)	Sector	IT
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
Submitted by	Matilde Casado and Margarida Antunes		One of nine std. scenarios
Date	2015		User defined distributions
File Name	C2015-21 Investment Product (1)	Project Control	Automatic tracking
			Tracking based on user input

1. Project description

Project authenticity

The project follows a structured lifecycle, covering supporting tasks, analysis, design, development, ICT testing, and implementation, ensuring thorough validation and testing before a full handover and go-live.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	17
Planned Duration (PD)	37 days*
Budget At Completion (BAC)	4.020 €
Renewable Resources	8
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	18%
Activity Distribution (AD)	35%
Length of Arcs (LA)	2%
Topological Float (TF)	36%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	17.5	18.8	-0.6
CRI-rho	49.5	13.6	0.9
CRI-tau	48.7	44.6	-0.3

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	38.6	11.7	2..7
CRI-rho	38.6	11.7	2..7
CRI-tau	38.6	11.7	2..7

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	19.2	36.2	-1.8
SI	32.6	31.5	-0.9
SSI	9.8	24.9	-2.6
CRI-r	12.9	20.6	-2.1
CRI-rho	33.6	22.7	0.1
CRI-tau	59.4	35.6	-0.3

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	1014.8	-1014.8	1	1.9	0.6
PV - SPI	940.9	-940.9	CPI	2.2	0.6
PV - SCI	933.9	-933.9	SPI	4.7	4.6
ED - 1	1659.2	-1659.2	SPI(t)	11.3	11.3
ED - SPI	940.9	-940.9	SCI	4.7	4.5
ED - SCI	934.3	-934.3	SCI(t)	11.2	11.2
ES - 1	1372.5	-1372.5	0.8 CPI + 0.2 SPI	2.5	1.7
ES - SPI(t)	242.0	-242.0	0.8 CPI + 0.2 SPI(t)	7.0	7.0
ES - SCI(t)	241.5	-241.5			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 2 tracking periods with a length of approximately 1 week. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	660.3	90.3	16.5	1.2	1.0	1.3	0.9
std dev	169.3	127.7	23.3	0.0	0.0	0.5	0.1
final	780.0	0.0	0.0	1.2	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	37 days	Real Duration	37 days	Late	0%
----	---------	---------------	---------	------	----

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	36.5	0.7	2.0	2.0
PV - SPI	37.5	0.7	0.7	0.7
PV - SCI	38.5	0.7	0.7	-0.7
ED - 1	39.5	0.7	2.0	-2.0
ED - SPI	40.0	0.0	2.6	-2.6
ED - SCI	40.0	0.0	2.6	-2.6
ES - 1	40.5	0.7	3.3	-3.3
ES - SPI(t)	41.5	0.7	3.6	-3.6
ES - SCI(t)	42.5	0.7	5.9	-5.9

2.3.3.3. Cost forecasting

BAC	4.020 €	Real Cost	3.240 €	Under Budget	-19.4%
-----	---------	-----------	---------	--------------	--------

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	3359.7	169.3	1.8	-1.8
CPI	3300.8	86.0	0.9	-0.9
SPI	3340.0	141.4	1.5	-1.5
SPI(t)	3213.2	38.0	0.4	0.4
SCI	3284.3	62.7	0.7	-0.7
SCI(t)	3178.2	87.3	1.0	1.0
0.8 CPI + 0.2 SPI	3307.9	96.0	1.0	-1.0
0.8 CPI + 0.2 SPI(t)	3277.9	53.6	0.6	-0.6